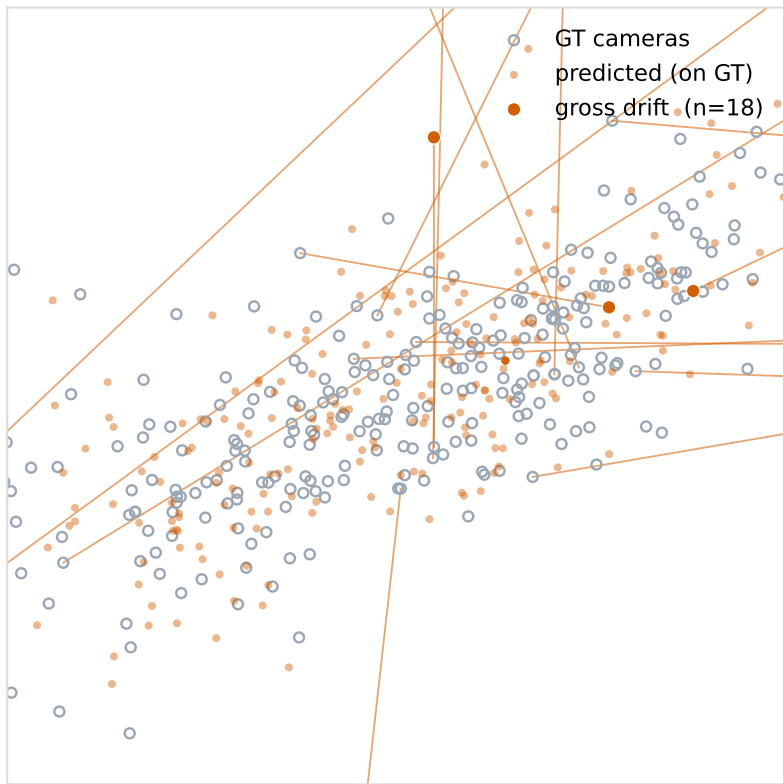
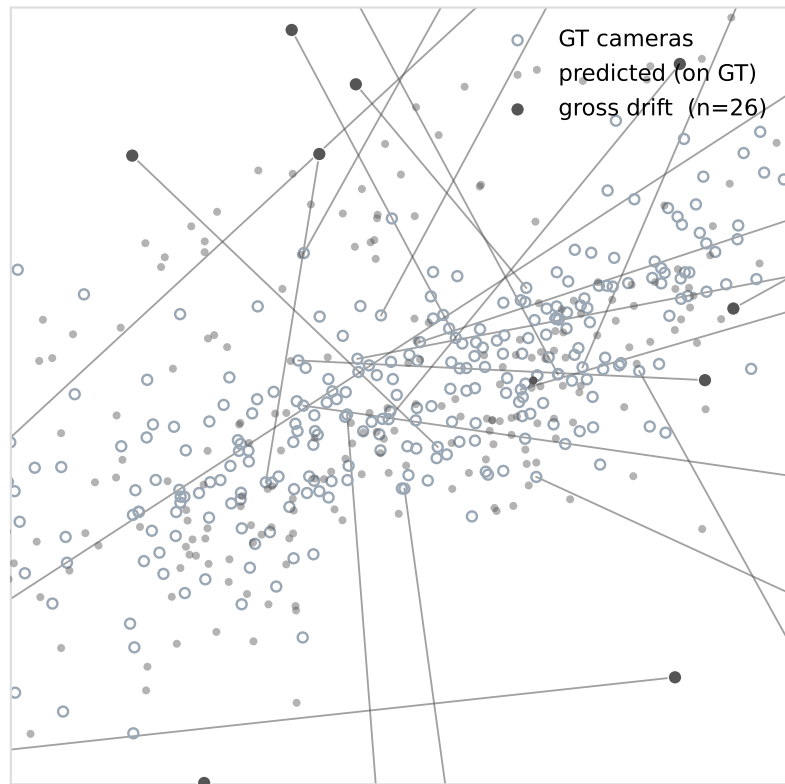


1DSfM · NYC_Library (32.6% outliers) — BA-aligned camera centers vs ground truth

madweight (ours, label-free)
translation error 1.32° · 18 gross-drift cams



RESfM (from-scratch, supervised)
translation error 5.42° · 26 gross-drift cams



Open markers = GT camera centers; small faint dots = well-placed estimates; large dots + connectors = gross-drift cameras (>0.5 -scene scale off GT). Label-free madweight leaves far fewer gross-drift cameras than the supervised from-scratch baseline. Cameras aligned to GT; same PCA view for both.